

4. Iron is extracted from iron oxide inside the blast furnace.



Iron oxide, coke and limestone are all fed in at the top of the furnace.

- (a) Describe the **two** main functions of coke in the furnace.

[3]

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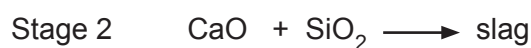
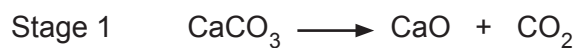
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- (b) Limestone is added to the furnace to remove some of the impurities as slag. Their removal involves a two-stage process as shown in the following equations.



- (i) A group of students were each asked to calculate the maximum mass of calcium oxide that could be produced from 8.0 g of calcium carbonate in stage 1.

The answers they calculated were as follows:

| Student | Answer |
|---------|--------|
| Yena    | 4.4 g  |
| Charlie | 4.5 g  |
| Tomasz  | 5.0 g  |

Give the name of the student that has correctly calculated the mass of calcium oxide to **one** decimal place.

You **must** show your working.

[3]

$$M_r(\text{CaCO}_3) = 100$$

$$M_r(\text{CaO}) = 56$$

Student .....



- (ii) In stage 2, the slag that is formed contains calcium silicate.

Calcium silicate is formed when calcium oxide (CaO) combines with silicon dioxide (SiO<sub>2</sub>) in a 1:1 ratio, with no other product formed.

Give the chemical formula of the calcium silicate formed in stage 2.

[1]

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- (c) Iron reacts with oxygen to form iron(III) oxide.

Complete and balance the equation for this reaction.

[2]

